

FULL-LENGTH ORIGINAL RESEARCH

Epilepsy surgery in children: Results and predictors of outcome on seizures

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SUMMARY

Purpose: To retrospectively analyze the results on seizures of surgery in children with drug-resistant focal epilepsy. To identify the factors predicting seizure control among several presurgical, surgical, and postsurgical variables.

Methods: One hundred thirteen patients (67 male, 46 female), younger than 16 years, operated on from 1996 to 2004 and followed-up for at least 2 years were identified. Individualized microsurgical resections, aimed at removal of the epileptogenic zone, were performed according to the results of tailored presurgical evaluations, which included stereo-electroencephalographic recording with intracerebral electrodes when needed. Risk of seizure recurrence was assessed for the considered variables by bivariate and multivariate analysis.

Results: Mean age at surgery was 8.8 years, mean duration of epilepsy was 5.7 years, and mean age at seizure onset was 3.1 years. One hundred eight patients (96%) had an abnormal magnetic resonance

imaging. At postoperative follow-up (mean duration 55.1 month), 77 patients (68%) were in Engel's class I, with 68 patients (60%) being seizure free (Engel's classes Ia and Ic). At multivariate analysis, variables associated with a significantly lower risk of seizure recurrence were unifocal lesion at MRI and older age at seizure onset (presurgical variables), temporal unilobar resection and complete lesionectomy (surgical variables), diagnosis of glial-neuronal tumors (postsurgical variables).

Conclusions: Surgery is a valuable option for children with drug-resistant focal epilepsies which may provide excellent results in a considerable amount of cases. Since results of surgery for epilepsy strongly depend on the presurgical identification of the Epileptogenic Zone, future work should be focused on refinement and implementation of diagnostic strategies.

KEY WORDS: Children, Epilepsy surgery, Drug-resistant focal epilepsy, Presurgical evaluation, Outcome.

In the last years, increasing consensus has grown on the efficacy of surgery to treat drug-resistant focal epilepsy in children (Wyllie et al., 1998; Leiphart et al., 2001; Bittar et al., 2002). In addition to control of disabling seizures, surgery may result in improvement of developmental, psychosocial and behavioural impairment experienced by children with early-onset epilepsy (Duchowny et al., 1998) and submitted to long-lasting antiepileptic therapy

(Loring and Meador, 2004). Moreover, early surgery in childhood may take advantage of the child's brain plasticity and enhance the chances of recovery from seizure-related damage and from possible postsurgical neurologic deficits. Nevertheless, the prognosis for seizure control is often a puzzling issue, and it may be determined by several variables. Furthermore, previous studies report controversial results concerning the factors which predict the postoperative outcome on seizures (Wyllie et al., 1998; Gashlan et al., 1999; Kim et al., 2000; Paolicchi et al., 2000; Kim et al., 2001; Kral et al., 2001; Leiphart et al., 2001; Sotero de Menezes et al., 2001; Kloss et al., 2002; Park et al., 2002; Porter et al., 2003; Hader et al., 2004; Hamiwka et al., 2005; Lee et al., 2005; Terra-Bustamante et al., 2005).

Accepted May 11, 2007; Online Early publication July 21, 2007.

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The aim of the present retrospective study was to analyze our experience in the surgical treatment of pediatric patients with drug resistant focal epilepsy, and to identify the factors predicting seizure control among several presurgical, surgical, and postsurgical variables.

PATIENTS AND METHODS

By retrospectively reviewing the records of the 492 patients operated on for medically refractory focal epilepsy at the “Claudio Munari” Epilepsy Surgery Center from May 1996 to November 2004, we identified 113 children (age at surgery under 16 years) with a postoperative follow-up of at least 2 years.

Presurgical investigations

Presurgical investigations and surgery were performed only after the patients' parents (or tutors) had given their informed consent. A comprehensive presurgical work up included: (1) history to establish age of onset, type and frequency of seizures; (2) neurologic examination and, when feasible, neuropsychologic profile; (3) interictal scalp EEG and, when needed, intensive video-EEG (VEEG) monitoring with at least one ictal recording; (4) brain magnetic resonance imaging (MRI) using appropriate sequences, with particular attention to the region(s) of presumed ictal onset.

When appropriate, invasive monitoring with stereotactically implanted multilead intracerebral electrodes (stereo-electroencephalography, SEEG) was performed. Indications, strategies of electrode implantation and results of SEEG in children have been already detailed elsewhere (Cossu et al., 2005). Briefly, SEEG is indicated when noninvasive investigations fail to satisfactorily localize the epileptogenic zone (EZ), according to one (or more) of the following patterns: (i) with a normal MRI, ictal/interictal scalp EEG findings are incongruous with ictal clinical semiology; (ii) with a focal MRI abnormality, electroclinical findings suggest a wide involvement also of extralesional areas; (iii) ictal clinical semiology is discordant with an apparently localizing ictal scalp EEG pattern (irrespective of MRI evidence); (iv) with large/diffuse/hemispheric/multifocal/bilateral MRI abnormalities, electroclinical evidences suggest a localized/lateralized ictal onset; (v) anatomical and/or electroclinical involvement of highly eloquent areas. In the latter instance, functional mapping conducted by intracerebral electrical stimulations (IES) allows identification of both eloquent cortex and subcortical critical bundles. Depending on the child's cooperation, IES are employed to identify primary motor, somatosensory, visual and speech areas (Cossu et al., 2005).

Surgery

All the patients received tailored microsurgical resections aimed at removal of the EZ, as defined by anatomoelectroclinical data. MRI-based neuronavigation (STP4.0,

Leibinger/Fischer, Freiburg, Germany) was employed in all operations.

Histology

Surgical specimens were routinely processed for histologic and immunohistochemical investigations. For histopathologic categorization, the revised WHO classification for tumors of the central nervous system (Kleihues and Cavenee, 2000) and a recent classification of focal cortical dysplasias (FCD) (Tassi et al., 2002) were adopted. In case of multiple histologic findings in the same patient, each pathology was considered with the same hierarchical value. Diagnoses resulting from previous resections performed at other Institutions, when available, were considered for subsequent statistical analysis.

Outcome on seizures and statistical analysis

Outcome on seizures was assessed according to the Engel's classification (Engel et al., 1993). For reoperated patients, the outcome referred to the last procedure.

Statistical analysis was performed to investigate the variability of seizure outcome, categorized as a dichotomous variable: absence (Engel's class I) versus recurrence (Engel's classes II to IV) of disabling seizures.

Variables included in the statistical analysis were: sex, neurologic status, age at seizure onset, duration of epilepsy, seizure frequency, MRI findings, use of VEEG and SEEG, age at surgery, type of surgery, side and site of surgery, extent of lesion resection, histology of resected tissue, length of follow-up.

Since the considered variables did not differ between “children” (younger than 12 years) and “adolescents” (older than 12 years) at a preliminary exploration by bivariate analysis, splitting the series into age-based subgroups was not considered appropriate for statistical purposes.

The site of surgical resection was categorized as a multinomial variable: (i) temporal unilobar resections; (ii) frontal unilobar resections; (iii) temporal “plus” resections (temporal lobectomy extended to either the parietal, occipital, or frontal lobes); (iv) posterior resections (unilobar or multilobar occipital, parietal and posterior temporal resections); (v) resections including the central area (alone or associated to frontal or parietal excisions); (vi) wide multilobar resections (more than three lobes involved in the resection). Extent of lesion resection (complete or incomplete) was determined basing on operative reports, postoperative MRI, histology of resection margins.

Homogeneous pathologies were merged if a preliminary bivariate analysis showed no differences in predicting outcome on seizures. The resulting histologic groups were considered as binomial variables (presence or absence of a given diagnosis).

For bivariate analysis, Kruskal-Wallis rank sum test was used to analyze numerical variables, and Fisher's two-tailed exact test was performed to analyze categorical (binomial or multinomial) variables. For multivariate

analysis, three logistic regression models (with Wald test) were fitted to separately assess the association of, respectively, presurgical, surgical, and postsurgical variables with the outcome on seizures. The explanatory variables were selected according to the results of bivariate analysis and of best subsets technique, as well as to the main evidences of the literature. *p*-values of <0.05 were considered as evidence of findings not attributable to chance.

Statistical analysis was performed using R 2.4.0 [R Development Core Team (2006). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. ISBN 3-900051-07-0, <http://www.R-project.org>].

RESULTS

There were 67 male and 46 female patients. The mean age at surgery was 8.8 years (range 1–15, SD 4.3), the mean duration of epilepsy was 5.7 years (range, 0–14; SD, 3.8) and the mean age at seizure onset was 3.1 years (range, 0–15, SD, 3.4). Twenty-one patients had a focal neurologic deficit at physical examination: 17 presented a lateralized motor impairment of variable severity, 2 a visual field defect, and 2 had both a motor impairment and a visual field defect. Epileptologic antecedents were reported in 11 cases, and consisted of threatened abortion in 4 cases, placenta detachment in 2 cases, fetal suffering combined with febrile convulsions in 2 cases, gestosis in 1 case, mother's abdominal trauma during pregnancy in 1 case. All patients experienced seizures despite adequate trials of at least 2 antiepileptic drugs appropriate for the seizure type without success. Seizure frequency was daily (more than 30/month) in 65 patients, weekly (five–30/month) in 29 patients, monthly (one–four/month) in seven patients, sporadic (less than one/month) in 12 patients. Eight patients underwent surgery in other Institutions before referral to our Center; in these cases, previous resection had aimed at removal of an identified lesion, with unsatisfactory results on seizure control.

Magnetic resonance imaging

In 108 cases (96%) brain MRI revealed anatomical abnormalities (89 unifocal, 10 multifocal and nine hemispheric). In five patients MRI was normal. Mesial temporal sclerosis (MTS) was suspected at MRI in seven cases (six with a focal lesion and one with a hemispheric abnormality).

Electrophysiologic investigations

Seventeen patients underwent only interictal scalp EEG evaluation. In these cases, full convergence, as to localization of the EZ, of interictal EEG abnormalities, ictal clinical semiology and MRI findings enabled surgery without further investigations. In 55 cases, scalp video-EEG recording of habitual seizures was required, along with MRI, to define the surgical strategy. The remaining 41

Table 1. Sites of surgical resections

Site of resection	No. cases	%
Temporal unilobar	43	38
Frontal unilobar	32	28
Temporal +	4	4
Parietal	2	
Occipital	1	
Frontal	1	
Posterior	20	18
Occipital	1	
Parietal	3	
Occipital-parietal	5	
Occipital and/or parietal + posterior temporal	11	
Including central area	9	8
Central alone	1	
Frontocentral	6	
Parietocentral	2	
Wide multilobar	5	4

patients received a SEEG investigation. Electrodes were stereotactically implanted monolaterally in 37 patients; four other patients had a prevailing monolateral implant, with some electrodes sampling also contralateral structures. The average monitoring period was eight days (range 1–21), and the whole procedure, including the sessions of IES for seizure induction and functional mapping, was generally well tolerated. The only complication was the breakage of an electrode under the cranial vault during an agitated seizure in a 13-year-old boy, which required surgical removal with no further sequelae.

Surgery

Five patients (all with a negative MRI) underwent a cortical resection (corticectomy), based only on the electroclinical findings (1 scalp video-EEG and 4 SEEGs). In 90 cases a cortical resection was associated to lesionectomy, and the remaining 18 patients received a pure lesionectomy.

Table 1 details the sites of surgery. Twenty-one unilobar (18 extratemporal, three temporal), and 20 multilobar resections were performed after a SEEG investigation.

Ten patients (2 temporal, 2 frontal, and 6 multilobar resections) underwent a second operation, owing to poor seizure control following the first resection.

Surgical complications were observed in 4 cases (1 epidural empyema, one depressed skull fracture caused by the Mayfield clamp, 2 transient supratentorial hydrocephali).

New permanent postoperative deficits occurred in 11 cases (10%). One of these (a 1-year-old female patient with catastrophic epilepsy) presented an expected left hemiparesis after resection of a right parietocentral Taylor's FCD; the other 10 patients had new visual field defects. New transient morbidity was observed in 15 cases (13 limb motor impairments, 1 central facial palsy, 1 diplopia).

Table 2. Pathologic findings disclosed at histologic examination of resected specimens

Diagnosis	No. cases
MCD	94
Focal cortical dysplasias	
Architectural	18
Cytoarchitectural	13
Taylor's type	25
Tuberous sclerosis	11
Other MCD	27
Tumors	43
Neuronal/glia-neuronal tumors	
DNT	16
Ganglioglioma	16
Gangliocytoma	2 (+1*)
Other tumors	7 (+1*)
Other pathologies	13
Chronic/Rasmussen encephalitis	5
Cavernous angioma	2 (+1*)
Infarction gliosis	3
Postoperative gliosis	2
Mesial temporal sclerosis	11
No histological abnormalities	2

The total number of pathologies exceeds the number of patients because 27 cases had multiple findings (see text).

*Diagnosis reported from other Institutions after previous surgery.

MCD, malformations of cortical development; WM, white matter; DNT, dysembryoplastic neuroepithelial tumor.

None of the 21 patients with preoperative focal deficits showed normalization of their neurologic status. In 3 cases, the preexisting deficits (2 hemiparesis, 1 visual field defect) were permanently worsened.

Histology

Results of histopathologic evaluations of surgical specimens are detailed in Table 2. Multiple pathologies were found in 27 cases. The most frequent association was among different forms of malformations of cortical development (MCD) (10 cases), followed by that of tumors with MCD (7 cases) and of MTS with other lesions ("dual pathology," 9 cases). In 1 case both postinfarction and postencephalitic changes were found. In 2 cases MTS was the sole finding. Three patients had no histologic changes, but in 1 of these a previous diagnosis of low-grade glioma after earlier surgery was reported from another Institution.

Outcome on seizures

All the patients had a postoperative follow-up of at least 24 months (mean, 55.1 months; SD, 24.8; range, 24–115). According to the Engel's classification, there were 68 (60%) seizure-free patients (Engel's classes Ia and Ic), 77 (68%) in class I, 10 (9%) in class II, 11 (10%) in class III and 15 (13%) in class IV. For one child deceased for malignant progression of a grade II ganglioglioma, the last avail-

able outpatient control (6 years postoperatively) was considered. The outcome did not significantly differ between patients younger or older than 12 years.

Statistical analysis

At bivariate analysis, the variables significantly associated with seizure outcome were (Tables 3 and 4): preoperative MRI ($p = 0.0009$), age at seizure onset ($p = 0.008$), complete lesion removal ($p = 0.0015$), site of surgical resection ($p = 0.00002$), histologic diagnosis of neuronal/glia-neuronal tumor ($p = 0.002$) and of encephalitis ($p = 0.035$).

The fitted regression models (Table 5) indicated a lower probability of seizure recurrence in patients with an unifocal lesion at MRI and in those with an older age at seizure onset (presurgical variables), in cases who received a temporal unilobar resection and in those submitted to complete lesionectomy (surgical variables) and in subjects with a diagnosis of neuronal/glia-neuronal tumors (postsurgical variables).

DISCUSSION

General considerations

The only selection criterion for this pediatric population was a postoperative follow-up of at least two years. Since it has been reported that 2-year postoperative outcome predicts long-term results on seizures (Salanova et al., 1999; Haniwka et al., 2005), it is reasonable to assume that results of surgery in our series are sufficiently stable.

The burden of disability experienced by children with drug-resistant focal epilepsy results from the combined effects of ongoing seizures and of antiepileptic medications. Frequently, a nearly complete control with only sporadic seizures is obtained at the expense of heavy medical therapies which may interfere with the child's developmental, cognitive, social and educational domains. Therefore, surgery should be considered also in children with sporadic seizures, as we did in 12 cases of our series, aiming at complete seizure control and freedom from medication.

Presurgical investigations

The present study indicates that excellent results may be achieved relying only on electroclinical information and brain MRI, without the recourse to functional imaging modalities like positron emission tomography or single-photon emission computed tomography.

MRI may provide substantial information which contributes to localize the EZ, especially when focal lesions are disclosed, as in most (79%) of our patients. When MRI is completely or partially uninformative, as in nonlesional cases and in those with extended or multiple abnormalities, the surgical strategy should be chiefly based on electroclinical investigations.

Invasive explorations with SEEG were required in 41 cases (36%). The different reported proportions of invasive

Table 3. Categorical variables and outcome on seizures: results of the bivariate statistical analysis (two-tailed Fischer's exact test)

Variable	Type	Categories	Frequencies (no. cases)	Engel's Class (%)		p-value
				I	II-IV	
Presurgical variables						
Sex	Binomial	male	67	72	28	0.412
		female	46	63	37	
Seizure frequency	Multinomial	sporadic	12	92	8	0.310
		monthly	7	71	29	
		weekly	29	66	34	
		daily	65	65	35	
VEEG	Binomial	performed	96	65	35	0.087
		not performed	17	88	12	
SEEG	Binomial	performed	41	59	41	0.141
		not performed	72	74	26	
Lesion at MRI	Binomial	unifocal	89	76	24	0.0009*
		not unifocal (multifocal/ hemispheric/absent)	24	38	62	
Surgical variables						
Complete lesionectomy	Binomial	performed	72	79	21	0.0015*
		not performed	41	49	51	
Resection side	Binomial	right	57	70	30	0.689
		left	56	66	34	
Resection site	Multinomial	Temporal	43	91	9	0.00002*
		Frontal	32	63	37	
		Posterior	20	60	40	
		Incl. Central	9	56	44	
		Temporal-plus	4	25	75	
		Wide multilobar	5	0	100	
Postsurgical variables						
FCD	Binomial	present	54	61	39	0.158
		absent	59	75	25	
Other MCD	Binomial	present	34	62	38	0.382
		absent	79	71	29	
Neuronal/glial-neuronal tumors	Binomial	present	35	89	11	0.002*
		absent	78	59	41	
Other tumors	Binomial	present	8	75	25	1
		absent	105	68	32	
Cavernous angiomas	Binomial	present	3	100	0	0.550
		absent	110	67	33	
Encephalitis	Binomial	present	5	20	80	0.035*
		absent	108	70	30	
Gliosis	Binomial	present	5	60	40	0.653
		absent	108	69	31	
Mesial temporal sclerosis	Binomial	present	11	64	36	0.742
		absent	102	69	31	
Histologic abnormalities	Binomial	present	111	68	32	1
		absent	2	100	0	

* Statistically significant.

VEEG, scalp video-EEG; SEEG, stereo-electroencephalography; FCD, focal cortical dysplasia; MCD, malformation of cortical development.

recordings, ranging from 24% to 73% (Wyllie et al., 1998; Kim et al., 2000; Paolicchi et al., 2000; Kral et al., 2001; Sotero de Menezes et al., 2001; Kloss et al., 2002; Porter et al., 2003; Hader et al., 2004; Hamiwka et al., 2005; Terra-Bustamante et al., 2005) probably results from differences in the employed technique (subdural or intracerebral electrodes), in the team's experience and in the complexity of selected patients.

Surgery

Most of our patients (62%) received an extratemporal or a multilobar resection, as expected in a pediatric population. These procedures may represent up to 90% of surgeries in youngest children (Duchowny et al., 1998; Wyllie et al., 1998), and recourse to invasive investigations is not infrequent in these cases (Munari et al., 1999; Sinclair et al., 2004).

Table 4. Numerical variables and outcome on seizure: results of the bivariate statistical analysis (Kruskal-Wallis rank sum test)

Variable	All patients median (IQ range)	Engel's Class		p-value
		I median (IQ range)	II-IV median (IQ range)	
Age at seizure onset (yr)	2 (1–4)	2 (1–5)	1 (0–3)	0.008*
Duration of epilepsy (yr)	5 (3–8)	5 (2–8)	6 (3–9)	0.352
Age at surgery (yr)	9 (6–12)	9 (6–12)	7 (6–12)	0.187
Follow-up duration (mo)	57 (32–72)	52 (32–63)	60.5 (34.3–79.5)	0.327

SD, standard deviation; IQ, interquartile; yr, years; mo, months; *statistically significant.

Table 5. Output of the fitted logistic regression models for, respectively, presurgical, surgical, and postsurgical variables, with recurrence of seizures (Engels classes II-IV) being the dependent variable

Variables	Reference category	Compared category	p-value	OR	CI _{OR}		RR	CI _{RR}	
					2.5%	97.5%		2.5%	97.5%
Presurgical (pseudo-R ² = 0.13)									
Preoperative MRI	Unifocal lesion	Not unifocal (multifocal/hemispheric/absent)	0.002	0.20	0.07	0.54	0.25	0.09	0.60
Age at seizure onset	–	–	0.03	0.83	0.69	0.97	NA	NA	NA
Surgical (pseudo-R ² = 0.23)									
Resection site	Unilobar temporal	Frontal	0.008	5.53	1.66	22.20	2.07	1.33	2.51
		Temporal-plus	0.012	25.78	2.51	615.72	1.18	1.18	1.33
		Posterior	0.013	5.85	1.52	25.78	1.99	1.26	2.36
		Incl. Central	0.039	6.10	1.09	36.05	1.88	1.05	2.20
		Wide multilobar	0.991	217836400	3.82e-133	NA	NA	NA	NA
Complete lesionectomy	No	Yes	0.05	0.40	0.23	1.01	0.45	0.18	1.01
Postsurgical (pseudo-R ² = 0.08)									
Neuronal/glia-neuronal tumor	Absent	Present	0.004	0.19	0.05	0.53	0.20	0.06	0.56
OR, odds ratio; CI _{OR} , confidence interval for OR; RR, risk ratio; CI _{RR} , confidence interval for RR; NA, not available.									

OR, odds ratio; CI_{OR}, confidence interval for OR; RR, risk ratio; CI_{RR}, confidence interval for RR; NA, not available.

Conversely, the proportion of unilobar temporal resections (38%) was lower, when compared with the adult series operated on in our Center in the same time frame (62%, data not shown). This figure is comparable to that of other studies (Paolicchi et al., 2000; Leiphart et al., 2001; Terra Bustamante et al., 2005), and series reporting more than 50% of temporal lobe cases had included patients as old as 18–20 years (Wyllie et al., 1998; Gashlan et al., 1999; Kim et al., 2000). Indeed, the diagnosis of temporal lobe epilepsy may be delayed until adolescence, because the electroclinical correlates of temporal lobe seizures are usually less localizing in younger children than in older patients (Holmes, 1986; Duchowny, 1987).

There was no formal hemispherectomy in this series. As a matter of fact, wide multilobar cases received sub-hemispheric resections, which spared as much as possible eloquent areas in patients with residual useful function, including 2 cases with Rasmussen encephalitis at a very early stage. Our presurgical work-up aims at identifying the area of seizure onset even in patients with extensive hemispheric damage and at sparing residual functions, if possible (Lo Russo et al., 2006). Nevertheless, in the light of the poor results obtained with this “conserva-

tive” approach, we now prefer complete hemispheric resections/disconnections in similar cases.

Concerning complications, only one case presented with a new, severe and permanent hemiparesis, indicating that new disabling morbidity after epilepsy surgery in a pediatric population may be limited to less than 1%.

Histology

The vast majority of pathologies were represented by MCD and tumors, a figure consistent with other major pediatric surgical series (Wyllie et al., 1998; Kim et al., 2000; Paolicchi et al., 2000; Leiphart et al., 2001; Terra-Bustamante et al., 2005). MTS was detected in one-fourth of hippocampal specimens, and it was associated with other lesions in most instances, indicating a prevalence of the so-called dual pathology in this population of children.

The presence of multiple pathologies in individual patients (24% in this series) may raise the question as to the respective etiological role of different coexisting substrates. Therefore, all the histologic findings were entered in the statistical procedure with the same hierarchical value, to investigate the impact of abnormal tissue

resection on seizure outcome, irrespective of the volumetric prevalence of one pathology on another.

Outcome on seizures

Patients with a unifocal anatomical abnormality at MRI (89% of cases in class I) had lower chances of seizure recurrence, compared with cases showing multifocal and hemispheric lesions or an unremarkable MRI. Similar findings have been reported in other pediatric series (Wyllie et al., 1998; Kim et al., 2000; Kral et al., 2001). The opportunity to disclose focal structural abnormalities offered by modern MRI may contribute to the excellent surgical results obtained in selected cases operated on without video-EEG recording, as in 17 of our patients. On the other hand, basing a surgical strategy exclusively on morphologic data would lead to disappointing results: as previously observed, the relationships between a presumably epileptogenic lesion and the EZ must be carefully ascertained (Munari et al., 2000). Although the ictal discharge may originate intralesionally, as well as in nearby cortical regions, the EZ may include cortical areas distant from the lesion (Munari et al., 2000). This issue is even more complex if one considers the possible interactions among site, extent, nature, and number of the structural abnormality (or abnormalities).

In the present series, age at onset of epilepsy influenced surgical outcome, with higher chances of recurrences being associated with earlier seizure onset. Recent studies on the effects of seizures in the immature brain may be useful to interpret this finding (Aamodt and Constantine-Paton, 1999; Stafstrom et al., 2000; Ben-Ari and Holmes, 2006). The establishment of an epileptogenic network at an early stage of brain maturation may result in a more widespread susceptibility to seizures, compared with brains experiencing the first seizure when the main physiologic networks have already developed. In such instances, focal resections could provide less satisfactory results because they interfere with only a part of more diffuse epileptogenic networks, possibly involving a larger amount of abnormally functioning cortex.

Significantly higher chances to achieve seizure control was observed in children submitted to unilobar temporal resections (with 91% of patients in class I), as compared with other sites of resection. This figure is consistent with previous reports on large pediatric series (Wyllie et al., 1998; Kim et al., 2000; Leiphart et al., 2001), and it can be ascribed to the more difficult localization of the EZ and to the need to spare eloquent areas in extratemporal cases. Unfortunately, although less gratifying, extratemporal resections are the most frequently performed operations in children with drug resistant epilepsy and further efforts, focused especially on the presurgical evaluation, are still needed to improve the surgical results in these more demanding cases.

Complete lesionectomy provided higher chances of a class I outcome (72%), as compared with incomplete lesion resection. Previous studies had indicated the same findings, irrespective of the pathology type (Paolicchi et al., 2000; Terra-Bustamante et al., 2005). Furthermore, complete lesion removal correlated with better seizure outcome either in children with tumor-associated epilepsy (Kim et al., 2001) or in those with cortical malformations (Kloss et al., 2002; Hader et al., 2004; Hamiwka et al., 2005). With most of the patients harbouring malformative and tumoral etiologies, our series confirms the high epileptogenic potential of these pathologies, whose complete removal is required to achieve control of seizures.

Significantly better results on seizures were obtained in patients with a glial-neuronal tumor (class I 89%) compared with other pathology types, as similarly shown by other studies (Wyllie et al., 1998; Kral et al., 2001). These maldevelopmental neoplasms are increasingly recognized as a cause of intractable epilepsy, especially of the temporal lobe, and the mechanisms underlying their intrinsic epileptogenicity are under investigation (Aronica et al., 2001).

The present retrospective study indicates that surgery is a valuable option to treat children with drug-resistant, often devastating, focal epilepsies, with 68% of patients being free from disabling seizures, and with an additional 19% significantly improved. These results were obtained relying on neuroradiologic (MRI) and electroclinical information. Major determinants of a favorable outcome in our population were a focal lesion at preoperative MRI, older age at seizure onset, an indication to a unilobar temporal resection, complete lesion removal and a tumor with a neuronal component as etiology. Since results of surgery for epilepsy strongly depend on presurgical identification of the EZ, future work should be focused on improving the contribution of available as well as of newly developed diagnostic tools to select surgical candidates.

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